

Heaven's Light is Our Guide



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Chapter 2 ~ Basic Relationship Between Pixels

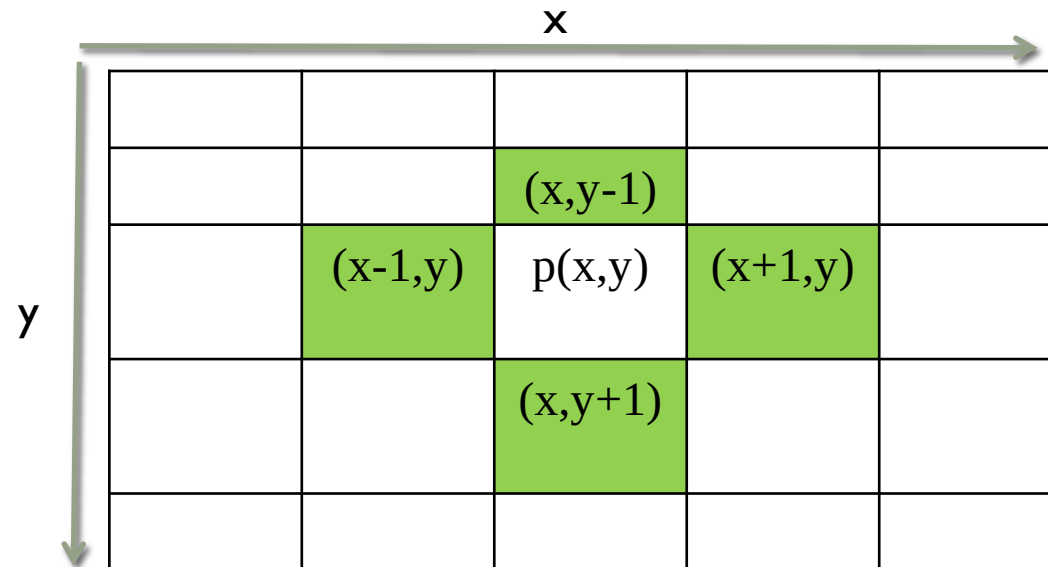
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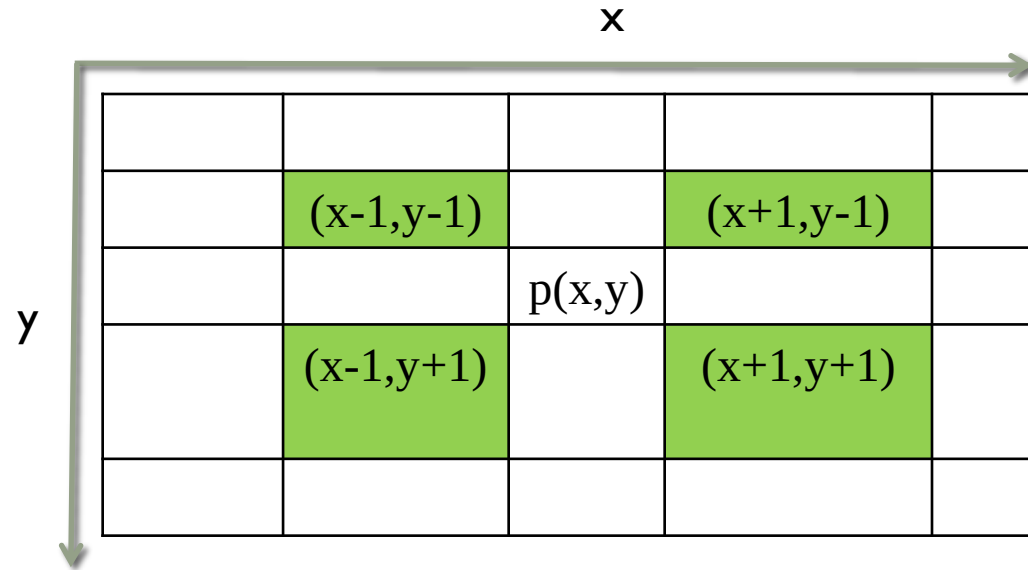
Neighbors of a pixel

- ✓ A pixel p at coordinates (x, y) has four horizontal and vertical neighbors.
- ✓ This set of pixels called 4 neighbors of p , is denoted by $N_4(p)$.
- ✓ Each pixel is unit distance from (x, y) .
- ✓ Some of the neighbor locations of p lie outside the digital image if (x, y) is on the border of the image.



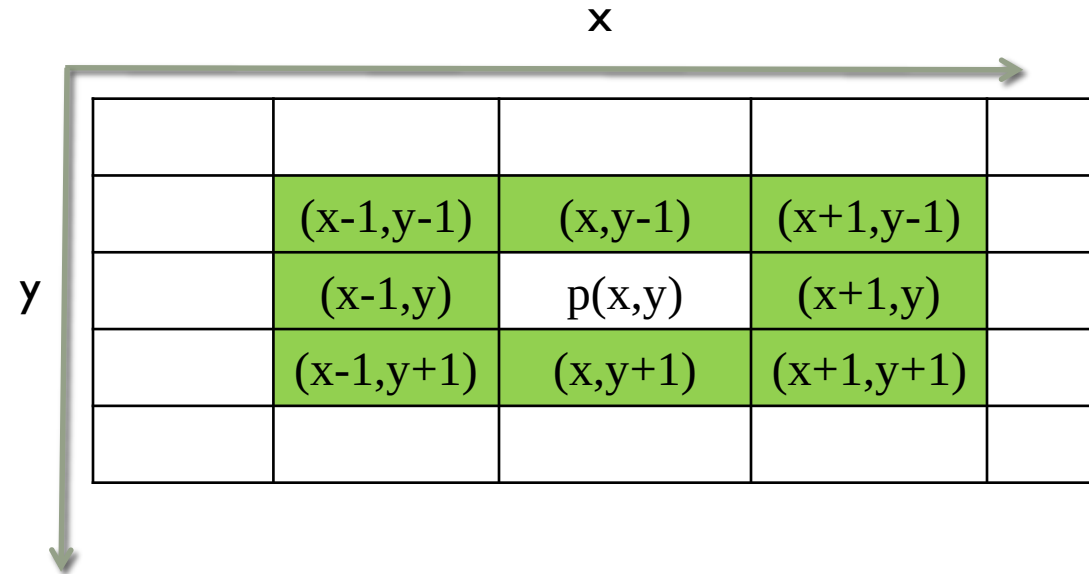
Neighbors of a Pixel

- ✓ The four **diagonal neighbors** of p at coordinates (x, y) is denoted by $N_D(p)$.



Neighbors of a Pixel

- ✓ The **8-neighbors** of p at coordinates (x,y) , denoted by $N_8(p)$.



Adjacency

- ✓ Two pixels are adjacent if they are neighbors and their intensity level 'V' satisfy some specific criteria of similarity.
- ✓ Let V be the set of intensity values used to define adjacency.
 - ✓ Binary Image : $V \subset \{0,1\}$
 - ✓ Gray Image : $V \subset \{0,1,2, \dots, 255\}$
- ✓ In binary images, 2 pixels are adjacent if they are neighbors & have some intensity values either 0 or 1.

Adjacency

Three types of adjacency:

✓ **4-adjacency:** Two pixels p and q with the values from set 'V' are 4-adjacent if q is in the set of $N4(p)$.

✓ For example, $V = \{0, 1\}$

1	1	0
1	1	0
1	0	1

✓ Here, p in **RED** color q can be any value in **GREEN** color.

Adjacency

Three types of adjacency:

✓ **8-adjacency:** Two pixels p and q with the values from set 'V' are 8-adjacent if q is in the set of $N_8(p)$.

✓ For example, $V = \{ 1, 2 \}$

0	1	1
0	2	0
0	0	1

✓ p in **RED** color and q can be any value in **GREEN** color.

Adjacency

Three types of adjacency:

m-adjacency: Two pixels p and q with the values from set 'V' are m-adjacent if

- (i) q is in $N_4(p)$ OR
- (ii) q is in $N_D(p)$ & the set $N_4(p) \cap N_4(q)$ have no pixels whose values are from 'V'.

For example, $V = \{1, 2\}$

0	1	1
0	2	0
0	0	1

Here, p in **RED** color and q can be any value in **GREEN** color.

Adjacency

m-adjacency: Two pixels p and q with the values from set 'V' are m-adjacent if

(i) q is in $N_4(p)$

For example, $V = \{ 1 \}$

(i) b & c

0 a	1 b	1 c
0 d	1 e	0 f
0 g	0 h	1 I

Solution: b & c are m-adjacent.

Adjacency

m-adjacency: Two pixels p and q with the values from set 'V' are m-adjacent if

(i) q is in $N_4(p)$

For example, $V = \{ 1 \}$

(ii) b & e

0 a 1 b 1 c

0 d 1 e 0 f

0 g 0 h 1 i

Solution: b & e are m-adjacent.

Adjacency

m-adjacency: Two pixels p and q with the values from set 'V' are m-adjacent if

- (i) q is in $N_D(p)$ & the set $N_4(p) \cap N_4(q)$ have no pixels whose values are from 'V'.

For example, $V = \{ 1 \}$

(iii) e & i

0 a 1 b 1 c

0 d 1 e 0 f

0 g 0 h 1 i

Solution: e & i are m-adjacent.

Adjacency

m-adjacency: Two pixels p and q with the values from set 'V' are m-adjacent if

- (i) q is in $N_4(p)$, OR
- (ii) q is in $N_D(p)$ & the set $N_4(p) \cap N_4(q)$ have no pixels whose values are from 'V'.

For example, $V = \{ 1 \}$

(iv) e & c

0 a 1 b 1 c

0 d 1 e 0 f

0 g 0 h 1 i

Solution: e & c are NOT m-adjacent.

Connectivity

Two pixels are said to be connected:

- ✓ If they are adjacent in some sense (neighbor pixels, 4/8/m-adjacency)
- ✓ If their gray levels satisfy a specified criterion of similarity (equal intensity level)

There are three types of connectivity on the basis of adjacency:

- 4-connectivity:** Two or more pixels are said to be 4-connected if they are 4-adjacent with each others.
- 8-connectivity:** Two or more pixels are said to be 8-connected if they are 8-adjacent with each others.
- m-connectivity:** Two or more pixels are said to be m-connected if they are m-adjacent with each others.

Paths

Example # 1: Consider the image segment shown in figure. Compute length of the *shortest-4*, *shortest-8* & *shortest-m paths* between pixels p & q where, $V = \{1, 2\}$.

	4	2	3	2	q
	3	3	1	3	
	2	3	2	2	
p	2	1	2	3	

Paths

Example # 1: Shortest 4-Path does not exist.

4	2	3		2 q
3	3	1		3
2	3	2		2
2 p	1	2		3



Paths

Example # 1: Shortest 8-Path = 4.

4	2	3	2 q
3	3	1	3
2	3	2	2
2 p	1	2	3

Paths

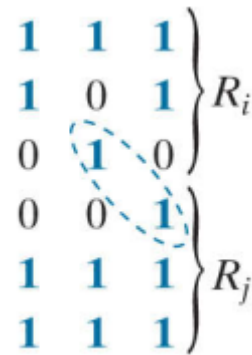
Example # 1: Shortest m-Path = 5.

4	2	3	2 q
3	3	1	3
2	3	2	2
2 p	1	2	3

The path highlighted by orange arrows is: $p \rightarrow 1 \rightarrow 2 \rightarrow 2 \rightarrow 1 \rightarrow q$. The total number of steps is 5.

Regions

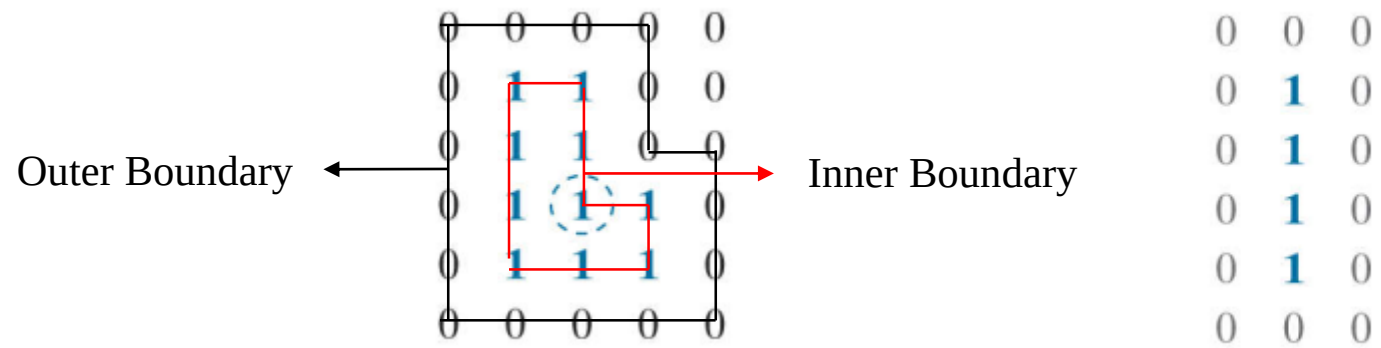
- ❖ A region in an image is a group of connected pixels with similar properties.
- ❖ Let R be a subset of pixels in an image. We call R a **region** of the image if R is a **connected set**.
- ❖ Regions that are **not adjacent** are said to be **disjoint**.



Two regions (of 1s) that are adjacent if **8-adjacency** is used.

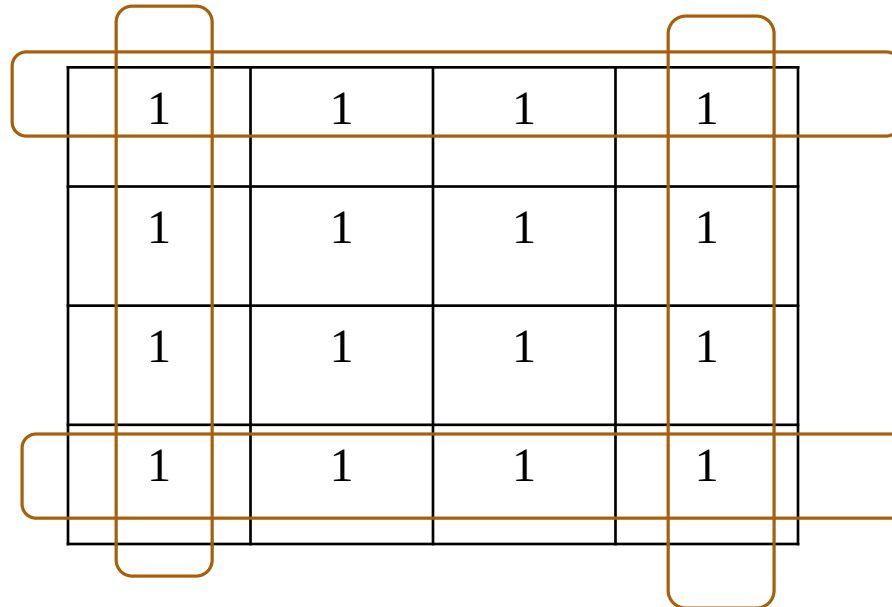
Boundary

- ❖ Set of pixels in the region that have one or more neighbors that are not in R.
- ❖ The boundary (also called the border or contour) of a region R is the set of pixels in R that are adjacent to pixels in the complement of R.
- ❖ Stated another way, the border of a region is the set of pixels in the region that have at least one background neighbor.



Boundary

- ❖ If R happens to be an **entire image** (which we recall is a rectangular set of pixels), then **its boundary** is defined as the set of pixels in **the first and last rows and columns of the image**.
- ❖ This extra definition is required because an image has **no neighbors beyond its border**.



Distance Measures

For pixels $p(x, y)$, $q(s, t)$ and $z(u, v)$, D is a distance function or metric if

- i. $D(p, q) \geq 0$
- ii. $(D(p, q) = 0, \text{ if } p = q)$
- iii. $D(p, q) = D(q, p)$
- iv. $D(p, z) \leq D(p, q) + D(q, z)$

	$p(x, y)$			
$z(u, v)$				$q(s, t)$

Euclidean Distance

❖ Euclidean distance between p and q is defined as,

$$D_e(p; q) = \sqrt{(x - s)^2 + (y - t)^2} = \sqrt{(2 - 4)^2 + (2 - 4)^2} = \sqrt{8}$$

	P(x, y)		
			q(s, t)

City Block Distance

❖ The city block distance D_4 between p and q is defined as,

$$D_4(p; q) = |x - s| + |y - t| = |2 - 4| + |2 - 4| = 4$$

	P(x, y)		
			q(s, t)

City Block Distance

??	??	2	??	=??
??	2	1	2	??
2	1	0	1	2
??	2	1	2	??
??	??	2	??	??

City Block Distance

4	3	2	3	4
3	2	1	2	3
2	1	0	1	2
3	2	1	2	3
4	3	2	3	4

Chess Board Distance

❖ The Chess board distance D_8 between p and q is defined as,

$$D_8(p; q) = \max|x - s|, |y - t| = |2 - 4| + |2 - 4| = 2$$

	P(x, y)		
			q(s, t)

Chess Board Distance

2	2	2	2	2
2	1	1	1	2
2	1	0	1	2
2	1	1	1	2
2	2	2	2	2

Thank You 😊